I built a production-grade AI credit-underwriting engine in a weekend.

Live. Dockerized. Unit-tested.
Here's the story ↓

A fintech PM portfolio piece

THE PROBLEM

B2B trade credit is still underwritten by hand.

30–90 min

per applicant — an analyst pulls financials, checks chargebacks, writes a memo.

Inconsistent

rigor drifts across analysts; no shared rule set.

Not auditable

justifications live in inboxes and spreadsheets.

3–5% losses

on extended credit when decisions get rushed.

WHAT I BUILT

One API call.
Full underwriting decision.
5 seconds.

Credit Score
0–100, deterministic

Credit Tier
A / B / C / D

Fraud Risk
Low / Medium / High

Cash-Flow Risk
From profit margin

Chargeback Risk
Count + rate

Recommendation
Approve / Review / Deny

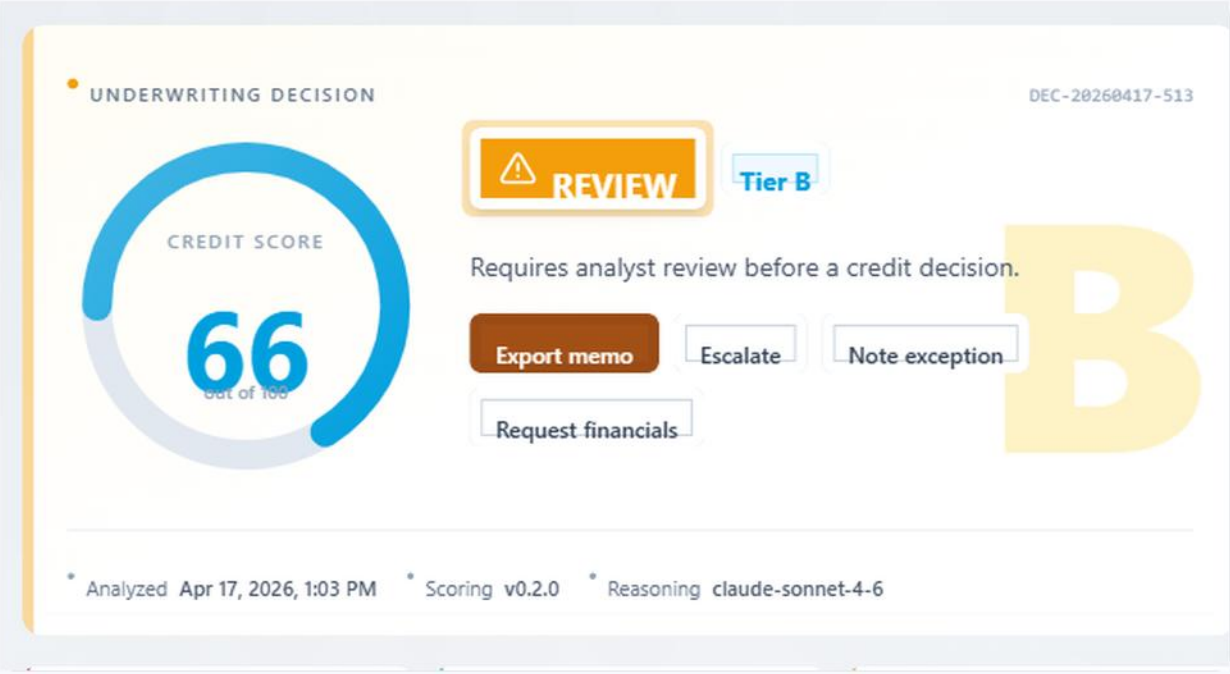
AI Memo
Claude Sonnet 4.6

Audit Trail
ID, timestamp, version

LIVE DASHBOARD

Analyst-grade decision UI.

Deployed on Render. Try it yourself.



Score 66 · Tier B · REVIEW (fraud override applied)

THE PRINCIPLE

LLMs don't underwrite. Rules do.

DETERMINISTIC

The score, tier, risks, and recommendation are pure functions.

22 unit tests cover thresholds, caps, and the fraud override rule.

Auditable. Testable.

LLM

Claude Sonnet 4.6 writes the underwriting memo — not the decision.

If the model fails, the app falls back to a deterministic narrative.

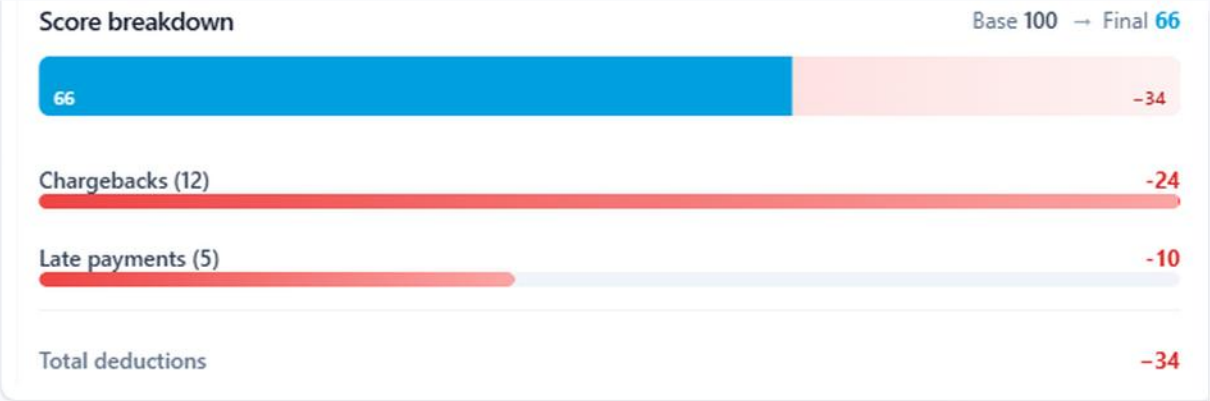
Explainable. Resilient.

Decisions that need to be right → code.
Decisions that need to sound right → LLM.

AUDITABLE

Every number is explainable.

Waterfall shows base 100 → final score, with each deduction visible.



Compliance and appeals get a defensible record.

Every decision carries a timestamp, scoring version, and model name.

PRODUCTION-READY

Not a demo.
A product.

22	unit tests on scoring + fraud rules
20/min	per-IP rate limiting on the API
UUID	request ID on every response + log line
CORS	origin allowlist via env var
React	error boundary — UI never blanks
Docker	single-origin deploy, Render auto-redeploy

SEE IT LIVE

Try the underwriting console yourself.

LIVE DEMO

credit-fraud-monitor.onrender.com

Click 'Run Risk Analysis' — results in 5s

SOURCE

github.com/mutamiri-sudo/ai-credit-underwriter

FULL PORTFOLIO

github.com/mutamiri-sudo/CharlesSokoTech

I'm open to Senior / AI PM roles in fintech, payments, and lending.

DM to chat. Repost if this resonates.